

Session 9

Technology Gaps, Innovation Systems, and Economic Complexity

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Technology gaps, innovation and complexity

1. Dependency theory as the vantage point
2. Technology gaps and innovation systems
3. The Product Space and Economic Complexity
4. Exercise with the Atlas of Economic Complexity
5. Case study: East Asia vs. Latin America
6. Possible critiques
7. Summary

01

What dependency theory leaves open

What dependency theory established

- Structural reproduction of underdevelopment – not a simple transitional state
- Core-periphery dynamics: development and underdevelopment linked
 - Unequal exchange, functional differentiation in the world system, colonialism
- International integration generates divergence, not convergence
- Policy implication: structural transformation requires breaking hierarchies

What dependency theory leaves unexplained

- How is the periphery's position reproduced at the level of firms and industries?
- Why does technology not diffuse automatically to poorer countries?
- Why do some economies break out of the periphery and others not — despite structural similarities?

“

Persistent international inequality is not merely a product of factor endowments or institutional quality, but of **structural differences in technological capabilities** that are self-reinforcing and path-dependent.

- Dependency theory and the technology-gap approach agree on outcome: **divergence**.
- They differ on the **mechanisms**: surplus extraction vs. absorptive capacity and relatedness.

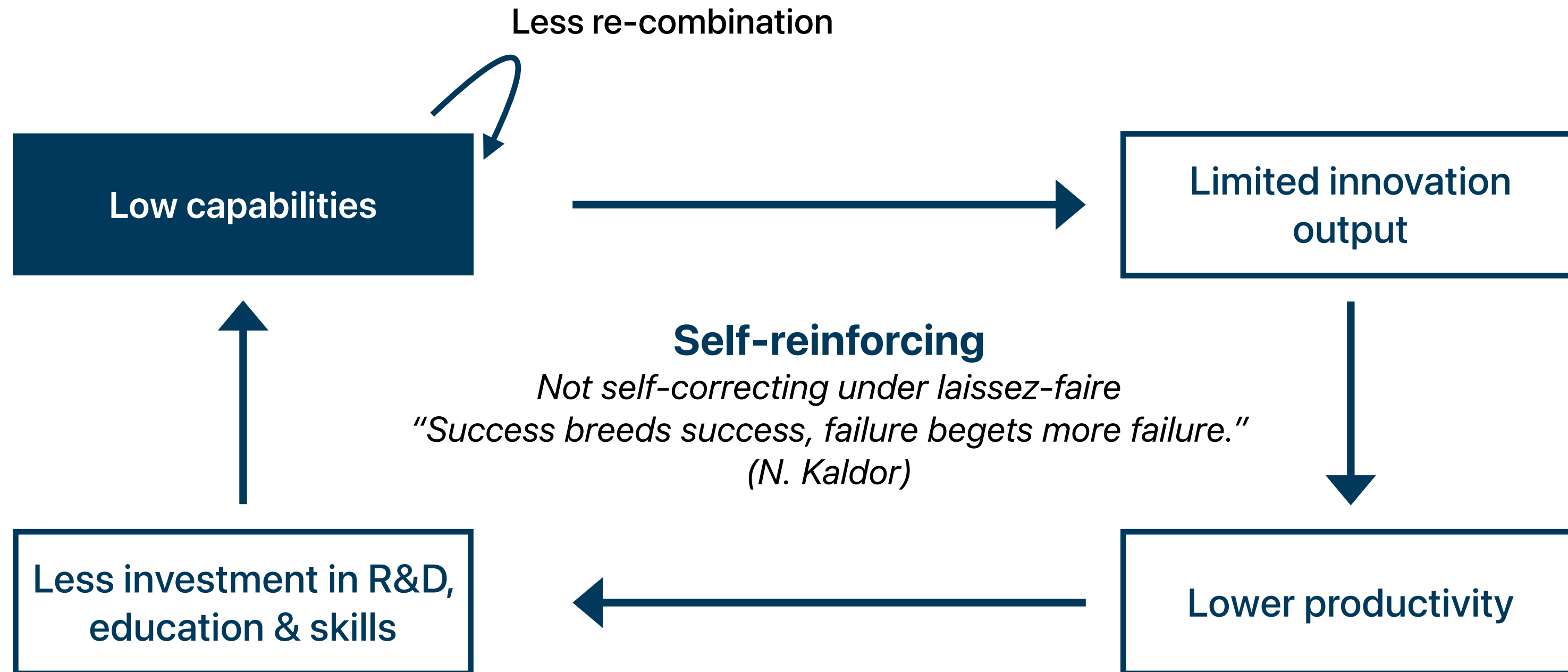
02

Technology gaps and innovation systems

The neoclassical baseline and why it fails

- Technology as non-rival, non excludable public good
 - Implies free distribution across borders
- Poorer countries may simply adopt existing technology
 - Convergence towards frontier without active technology policy
- Tacit knowledge
 - Capabilities embedded in routines, skills, ...
 - Cannot be fully codified and transferred
- Path dependence
 - Where you are → where can you go
 - Falling behind → capacity to catch-up ↓
- Absorptive capacity
 - Adopting techs requires prior capability

The capability trap



Competitiveness: costs or technology?

- Two competing theories of trade competitiveness
 - Neoclassical: competitiveness = relative **costs** → lower wages → more exports
 - Tech. Gap: competitiveness = absolute **technological** advantage → who can produce what
- Both relevant, but technology tends to dominate
 - **Kaldor paradox**: successful export nations with fastest growth in unit labour costs
 - Dosi et al. (2015, see readings) as recent evidence
- Evolutionary **micro- and meso-foundations**
 - Capabilities are tacit and sticky → focus on micro and meso level learning processes
 - Relevance of selection processes → **persistent heterogeneity** as normal state

Competitiveness: costs or technology?

- Two competing theories of trade competitiveness
- Both obviously relevant, but technology tends to dominant
- Evolutionary micro-foundations

Important policy implication

If technology drives long-run competitiveness, internal devaluation (wage cuts) is the wrong lever.

The relevant instrument is capability investment: R&D, education, and directed industrial policy.

National Innovation Systems (NIS)

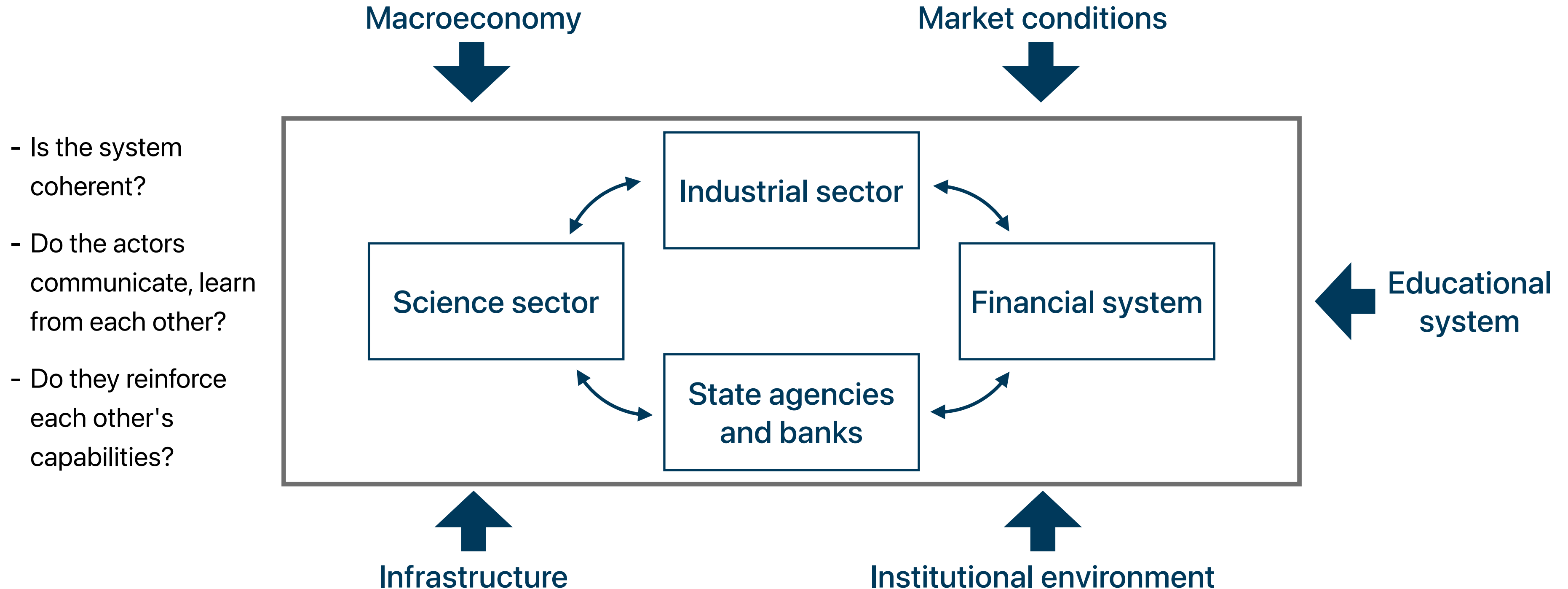
- Innovation as a systemic process
 - No single firm or policy lever substitutes the ecosystem

[...] the network of institutions in the public and private sectors whose activities and interactions initiate, import, modify and diffuse new technologies."

Christopher Freeman



National Innovation Systems (NIS)



Comparison with earlier frameworks

Framework	Shared with NIS	Key differences
New Institutional Economics	Institutions fundamental to development	Property rights and transactions costs less relevant
Marxism / Dependency theory	Global system yields divergence endogeneously	Learning and absorptive capacity instead of surplus extraction and unequal exchange
Neoclassical growth models	Capital and human-capital investment matter	NIS rejects free diffusion assumption; relevance of absorptive capacity

03

The Product Space and Economic Complexity

The key relevance of collective knowledge

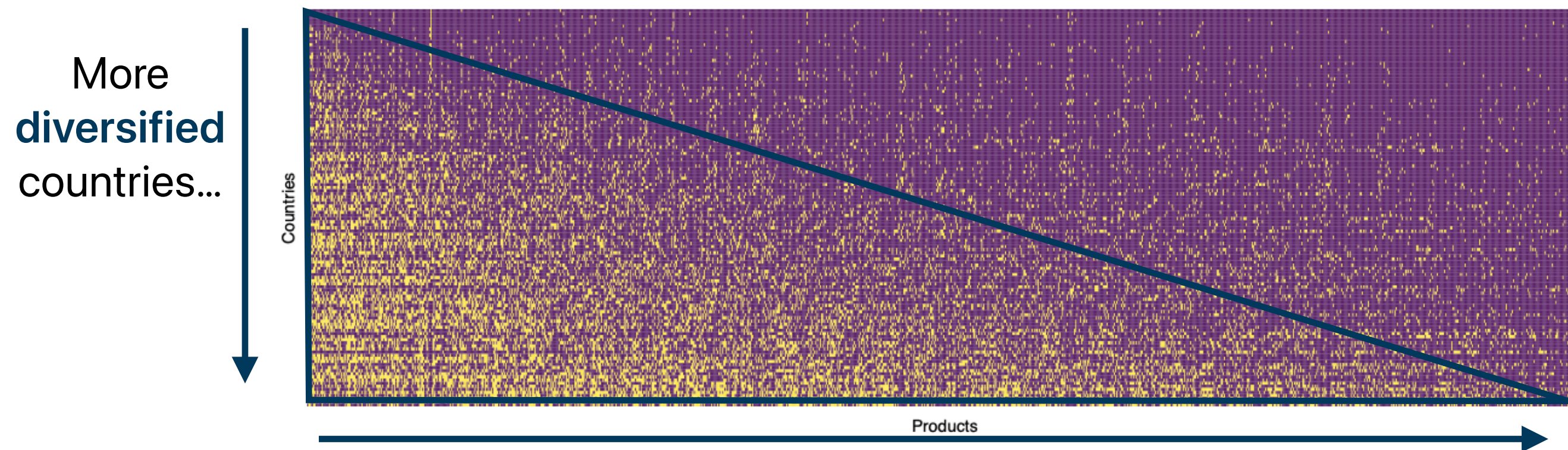
- Idea of **economic complexity**: collective knowledge as source of prosperity
 - Collective knowledge $\neq$ individual knowledge (a.k.a. a person byte)



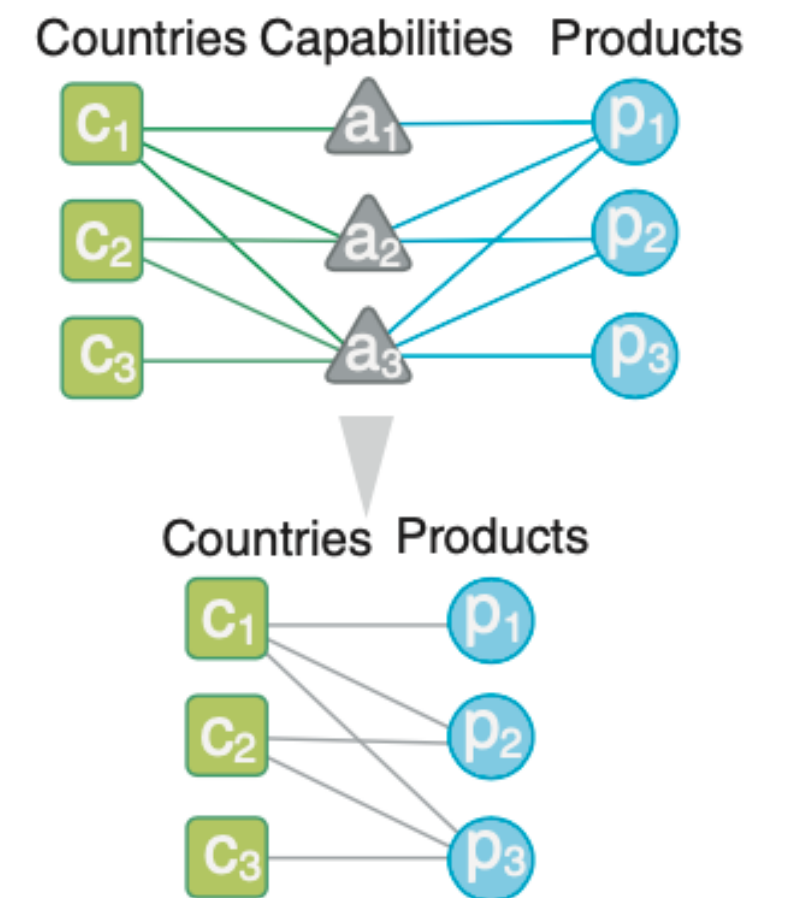
- Working together enables activities requiring many person bytes

Crystallised capabilities: economic complexity

- **Measurement innovation:** products as **crystallised information**
- **Stylized fact:** countries produce everything they can
 - More capabilities → more diversified production → more wealth

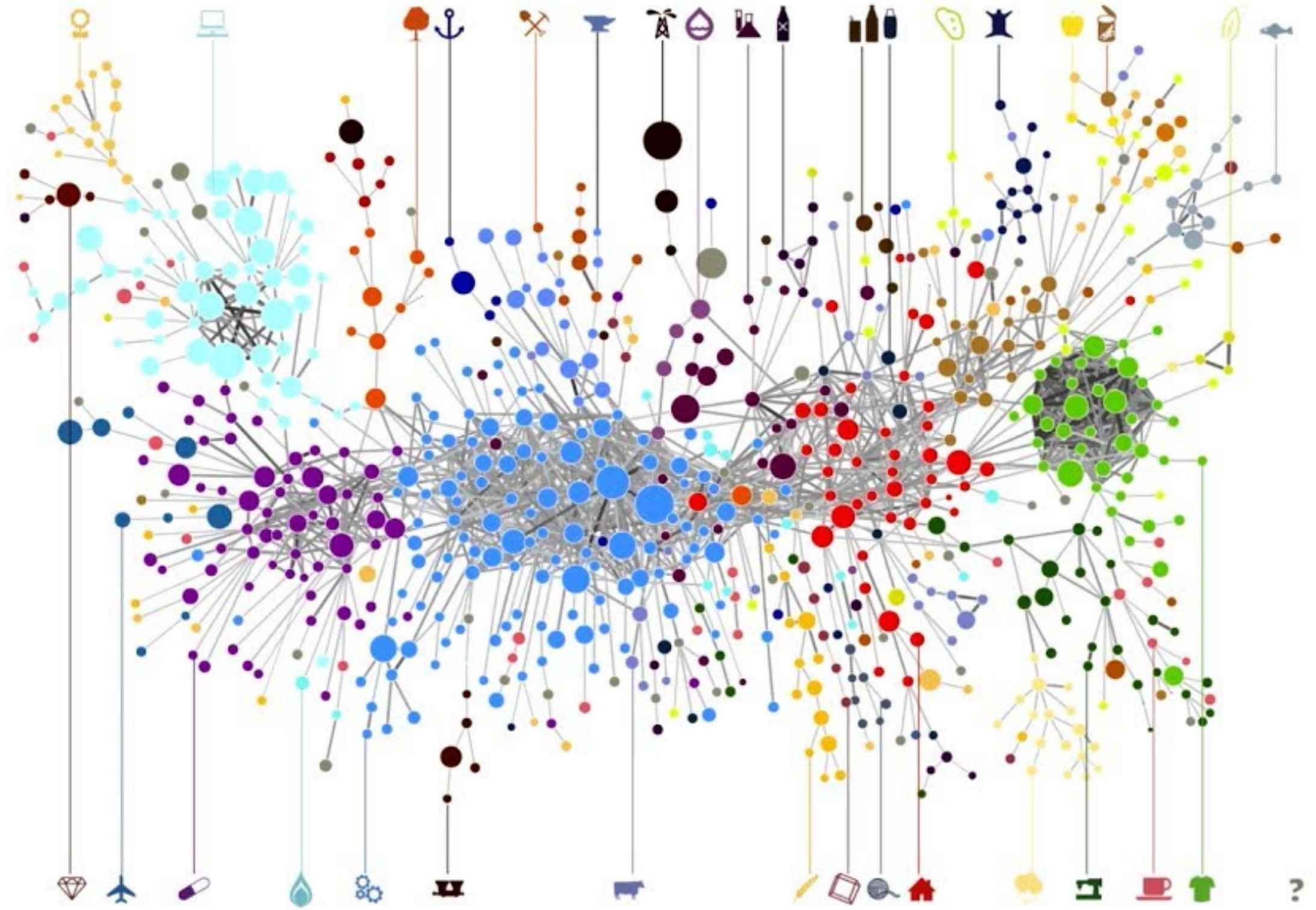


...also tend to produce less **ubiquitous** products.

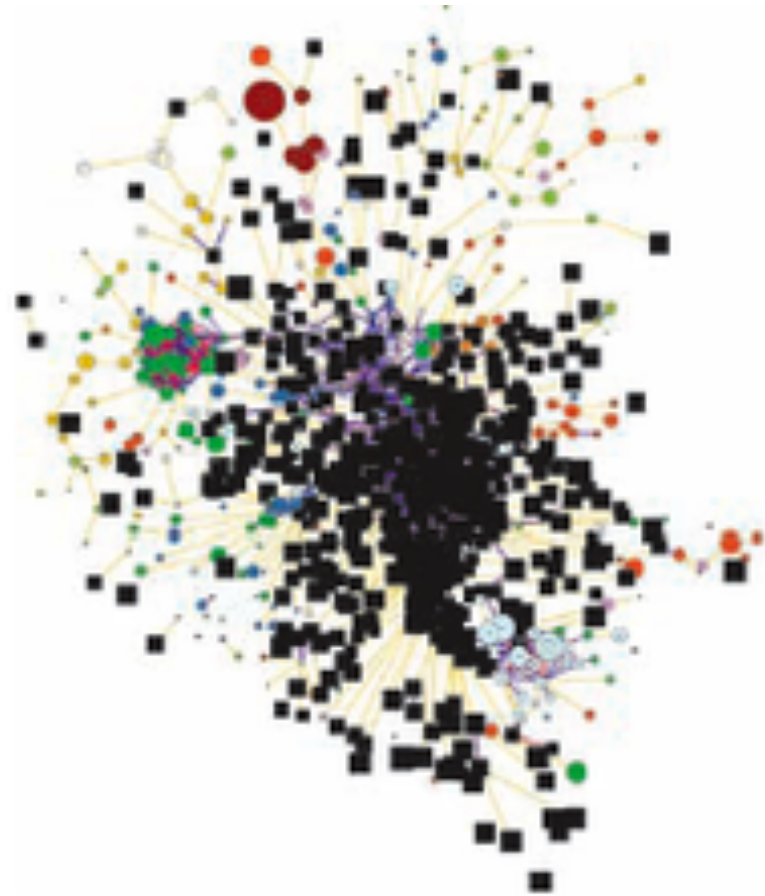


The product space

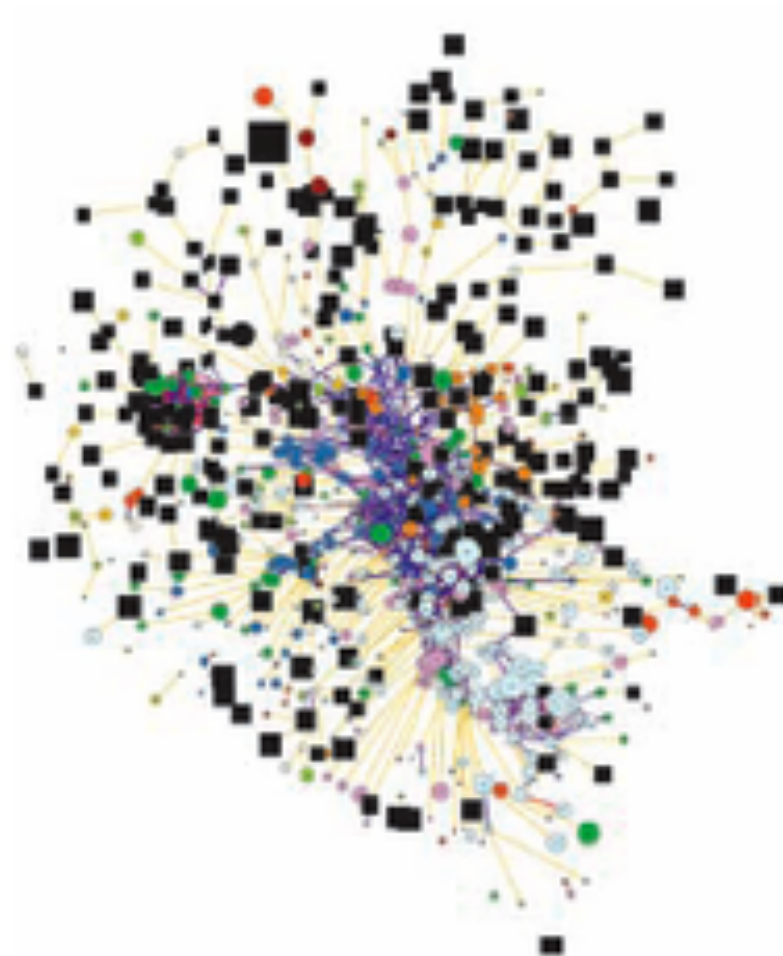
- **Products** as vertices
- Edges as **proximity**
 - Proximity: likelihood of being co-exported
 - Close products require similar capabilities
- Products in the **center**: the **most valuable capabilities**
 - High **Product Complexity Index (PCI)**



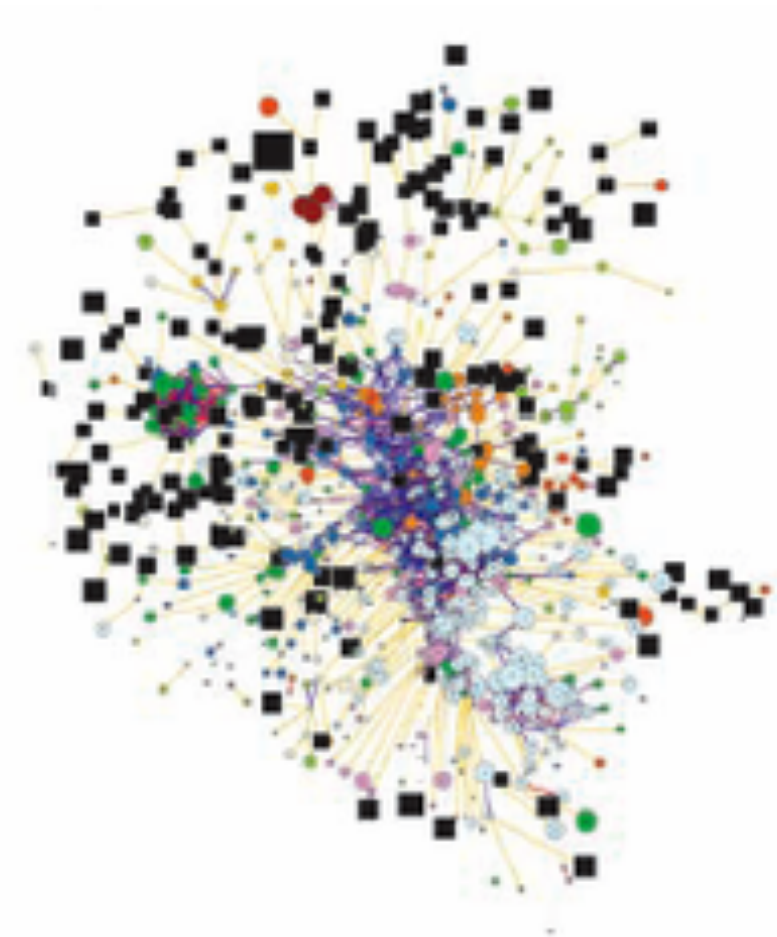
The product space - country examples



Industrialised
countries



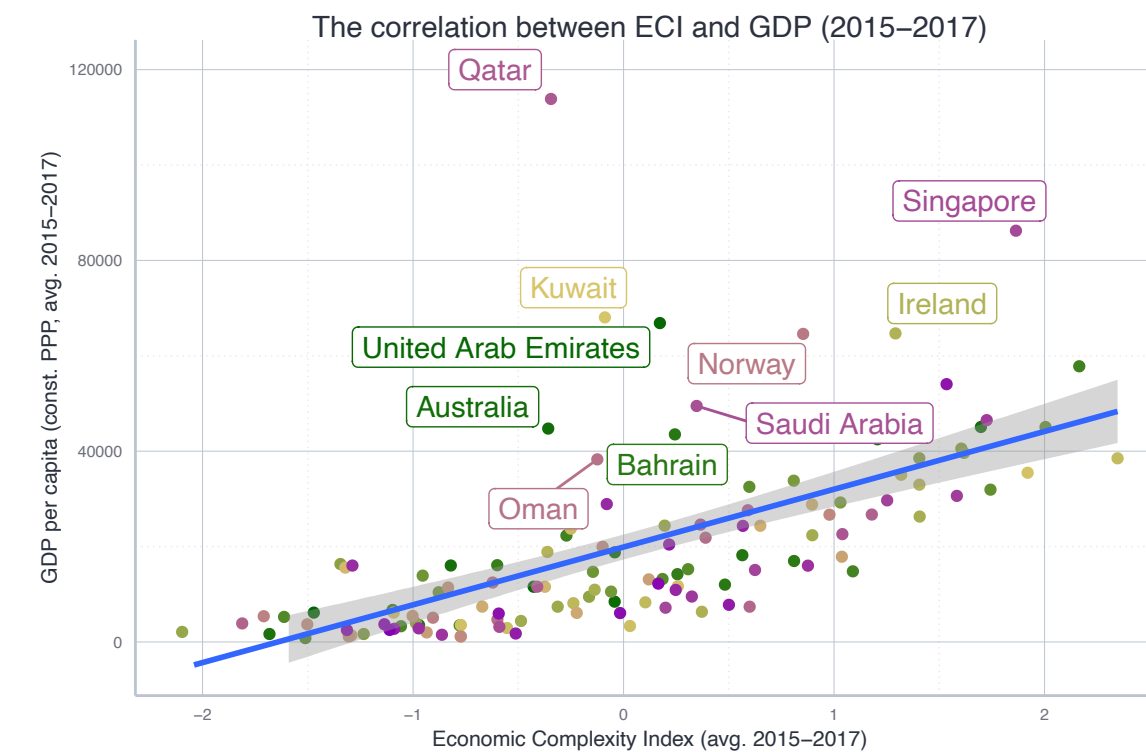
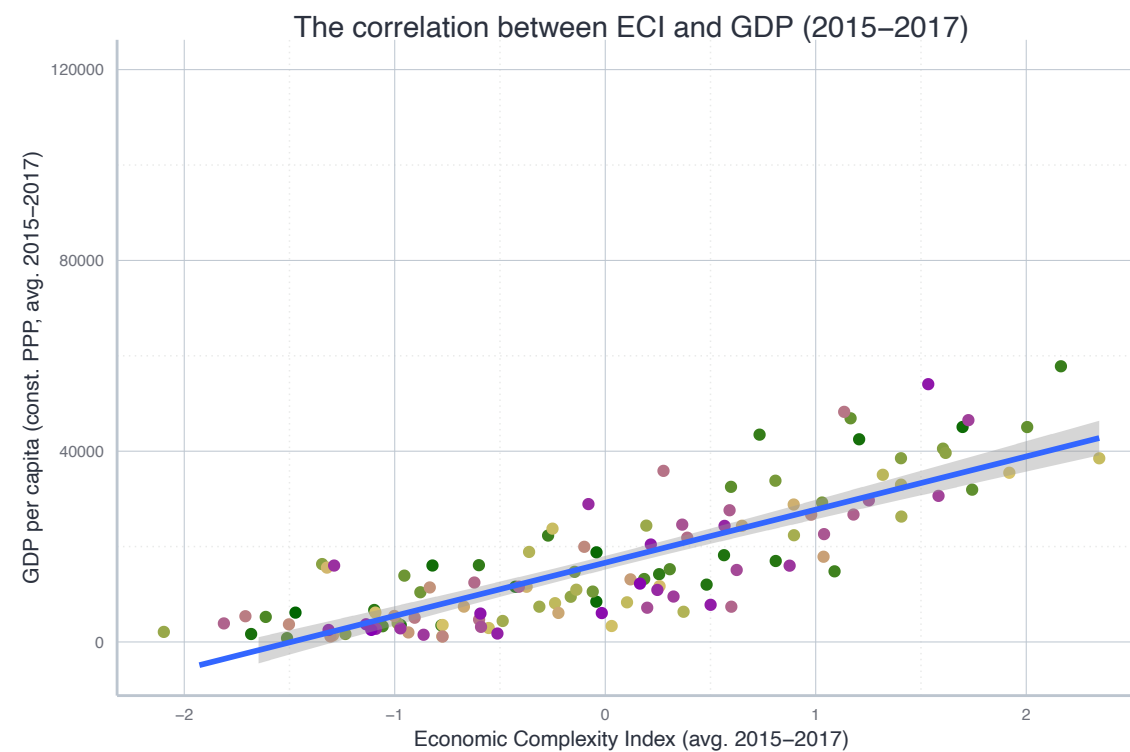
Latin American
countries



African
countries

The Economic Complexity Index

- Country pendant to the PCI
- Higher ECI → more collective capabilities → produce products with high PCI
 - It takes a complex country to make a complex product
 - Complex products are those produced (only) by complex countries

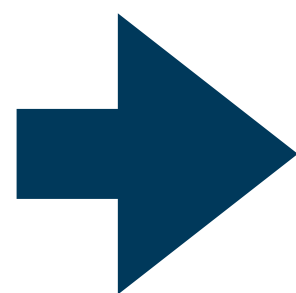


Capability accumulation and relatedness

- How to accumulate new capabilities and increase complexity?
 - Especially: capabilities to do good things (i.e. that make your country rich and keep it clean)



- People learn things similar to what they know → **relatedness**
- More capabilities → easier to accumulate more capabilities (**recombination**)



Strong path dependence in the context of capability accumulation!

The Atlas of Economic Complexity



HOME EXPLORE COUNTRIES DATA LEARN ABOUT

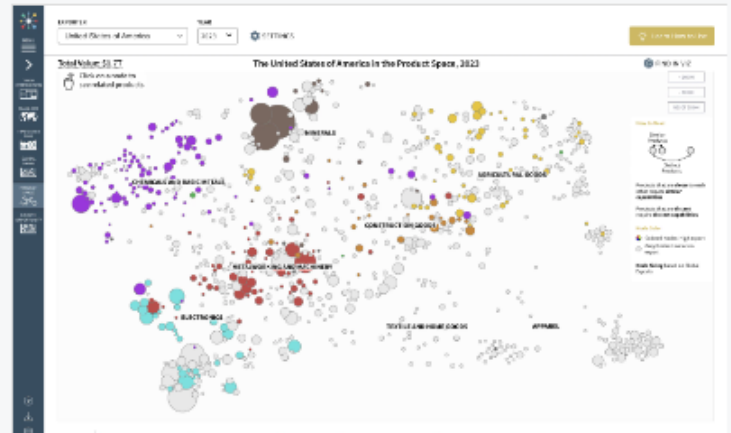
ATLAS OF ECONOMIC COMPLEXITY

Harvard Growth Lab's research and data visualization tool used to understand the economic dynamics and new growth opportunities for every country worldwide.



Rich resource for trade and development data → Exercise

TRADE DYNAMICS & GROWTH OPPORTUNITIES



Build custom data visualizations to reveal 50+ years of global trade flows and opportunities for growth. [Start Exploring →](#)

COUNTRY PROFILES



An interactive journey through a country's economic structure and growth patterns to uncover strategies for greater prosperity. [Start Exploring →](#)

04

Exercise: using the Atlas

Using the Atlas of Economic Complexity

- Choose a country of interest
 1. Investigate its contemporary trade composition → Tree map
 2. What are its most important trading partners? → Trade map
 3. How did both properties have changed since 1990? → Trade over time
 4. How is it positioned in the product space? → Network
 5. What do the growth opportunities analysis imply for this country? → Table & Scatter plot
- Consider this a first experimentation with a very rich resource!

05

Case Study: South Korea & Brazil

Two regions, two different development paths

- South Korea and Brazil as exemplars for Latin America and East Asia

1960

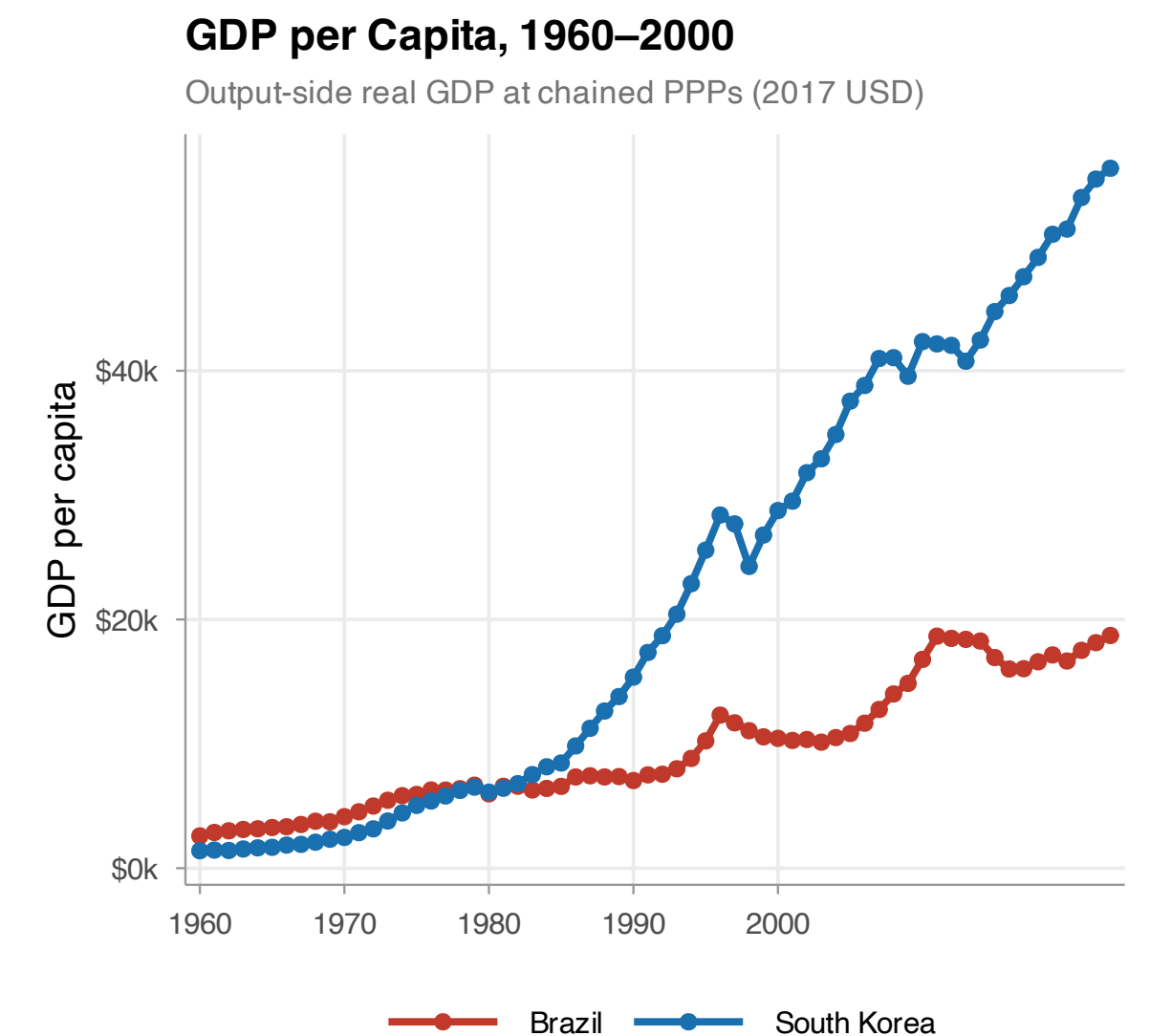
Comparable incomes,
industries, and constraints



2000

South Korea moved to core,
Brazil remains in periphery

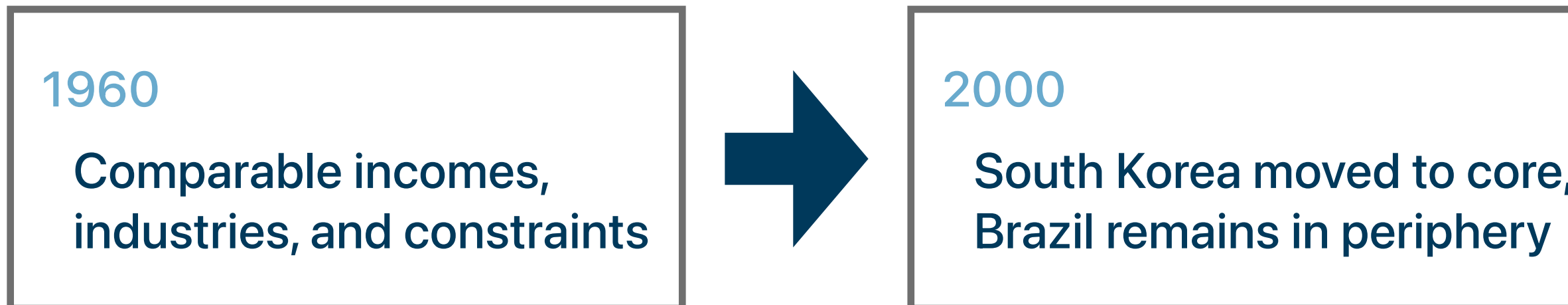
- What NIS highlights/shallows as compared to NIE and Dependency theory?



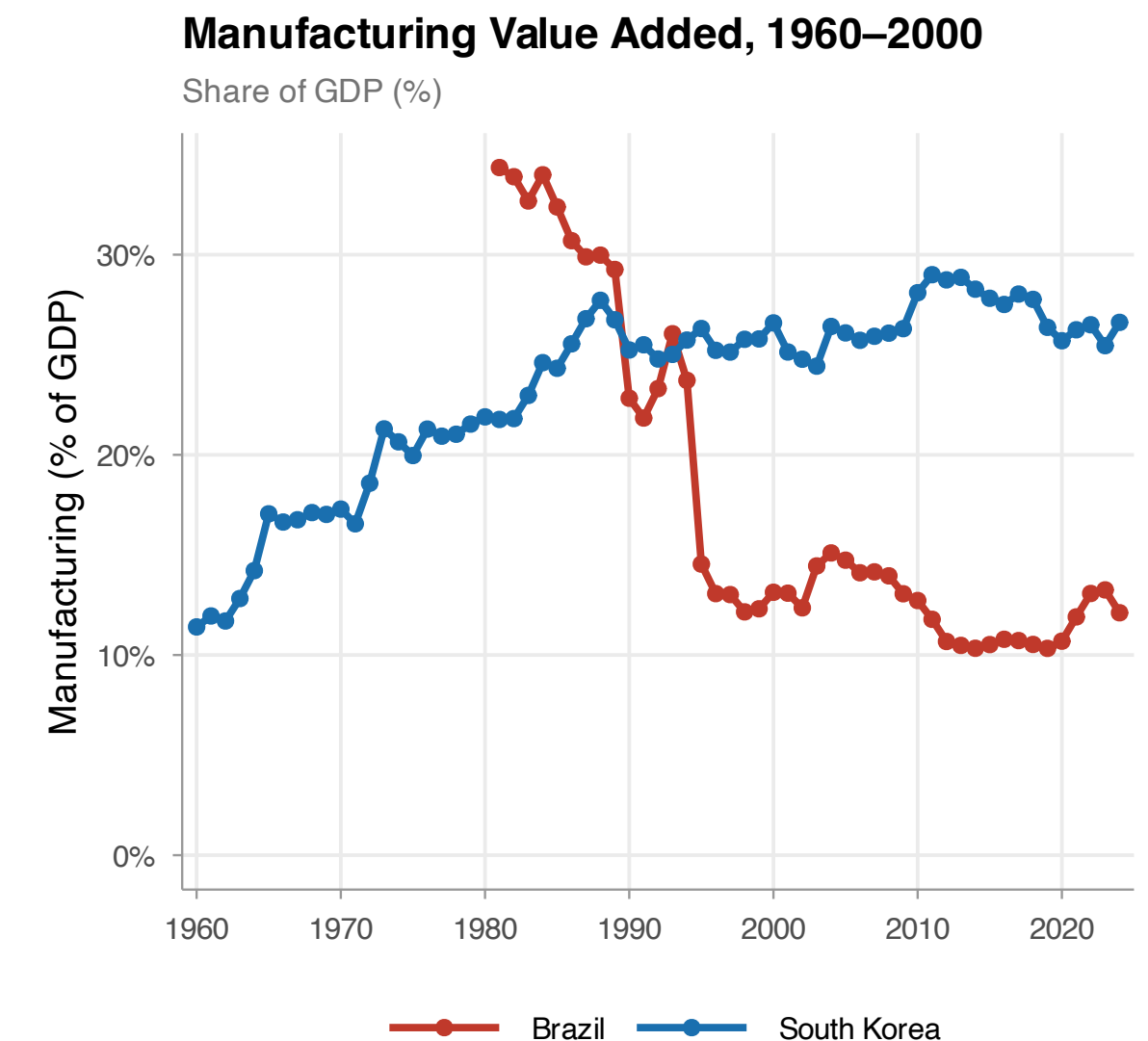
Source: Penn World Tables 11.0 (Feenstra, Inklaar & Timmer 2015). Variable: rgdpo / pop.

Two regions, two different development paths

- South Korea and Brazil as exemplars for Latin America and East Asia



- What NIS highlights/shallows as compared to NIE and Dependency theory?



Source: World Bank World Development Indicators. Indicator: NV.IND.MANF.ZS.

South Korea and its deliberate capability building

- **Directed credits** to strategic sectors via development banks
- **Infant-industry protection** – selective and time-limited
- **Technology-transfer** requirements on inward FDI
- Strong focus on **exports**

NIS diagnosis

- Coherent state-industry coordination
- Long time horizons, clear incentives
- Intervention against short-run markets signals
- Sequential, managed moves on product space

Brazil and the problems of Latin America

- Successful **import substitution** only until 1960s
- Continued dependence on **simple commodities**
- No technology learning from inward **FDI: enclaves** with weak linkages
- 1980 debt crises as **macro problem**
- **Political instability** in LA, short horizons

NIS diagnosis

- Lack of state-industry coordination
- Short time horizons, repeated policy reversals
- Stayed in sparse periphery of the product space
- Problematic infrastructure, turbulent political env.

An overall comparison

	East Asia	Latin America
State-industry coordination	Active developmental state	Rather weak, no coherent program
FDI strategy	Transfer requirements Domestic linkages	Resource enclave Limited spillovers
Political environment	Stable, massive aid and security guarantees by US	Highly volatile, conflict with US
Macroeconomic stability	Stable, long investment horizons	Volatile, debt crises
Product space trajectory	Deliberate, sequential move from periphery to core	No strategic moves, stuck in periphery

No paradigm is perfect

Technology gap approach

- Highlighting requirements for learning
- Role of university–industry–government linkages
- Ignorance of the **geo-political environment** and requirements
- **De-politisation** of innovation

Dependency Theory

- The role of **political environment** and external actors (US, IMF,...)
- The interests of local elites and barriers to integration
- **Marginalization of domestic agency** and micro processes in general
- **Over-determinism**

New Institutional Economics

- Relevance of **state capacity**
- Institutional **persistence**
- South Korean developmental state largely built on extractive institutions
- Technology and learning **black-boxed**

06

Critical discussion

Typical critiques of national innovation systems

1. Latent Eurocentrism
2. State capacity as prerequisite
3. Neglect of global power constraints
4. Selective evidence presentation

Critique 1: Latent Eurocentrism

- Central narrative: catching-up to the Global North
 - Modernisation theory in evolutionary clothing
 - Alternative notions of development not considered
- No reflection on exploitation as a prerequisite
- Little reflection on the social and ecological feasibility of such catching-up

Open question

To what extent can the framework accommodate alternative development paths?
Or is it structurally bound to the modernization theory?

Critique 2: State capacity as a prerequisite

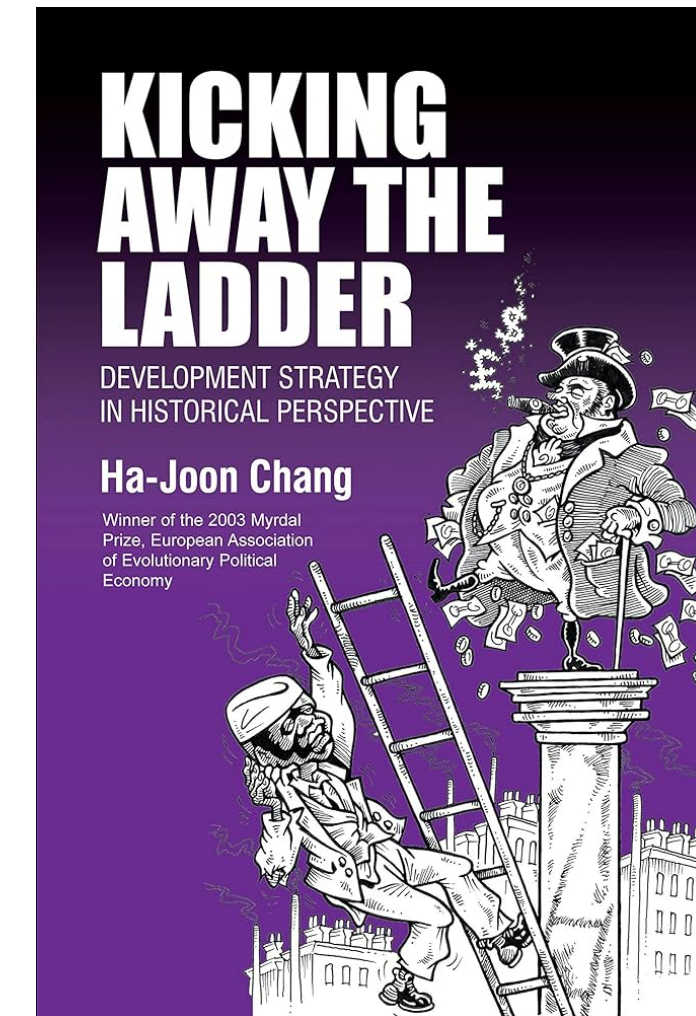
- Policy prescriptions *assume* capable state and developmental bureaucracies
 - A Korean EPB or Taiwanese ITRI equivalent
- No suggestion on how to build up these state capacities
 - Without such state, policies cannot be implemented
- No reflection on the required political environment

Note

State capacity as both something to be explained, as well as a tool for further policies.

Critique 3: Neglect of global power constraints

- East-Asia received huge support by the US
 - Security guarantees, market access, subsidies...
- Latin America suffered from continued external interventions
- Most successful policies of South Asia not banned under WTO rules



Hypothesis: NIS lacks the political economy lens

Early industrializing countries built capabilities under conditions they now (selectively) deny to others - for political reasons not reflected upon in the NIS.

Critique 4: Selective evidence presentation

- East Asia by far most researched case in the literature
 - But: very specific historical situation
- Much less attention on failures:
 - Brazil's IT protection policy in 70s and 80s
 - Failed policy experiments in Africa

Caution

The existing literature appears sloppy when it comes to generalizing insights from specific socio-historical situations.

07

Summary and outlook

Summary

- Comprehensive view on **capability accumulation** and its development implications
- Similar conclusions as dependency theory, but different mechanisms at centre stage
- Economic **complexity** and the **product space** as a contemporaneous framework
- Four potential weaknesses:
 - Eurocentrism, weak theory of the state (capacity) and political economy aspects, lack of socio-historical specificity
- In all: **important complement** to the other three frameworks so far